

Specification

Introduction

This is the specification of a quizlet-like mobile app named “Quizzeit.” “Quizzeit” is designed to help students enhance their understanding of high school and university-level Formal Logic concepts. The app provides a platform where users can test their knowledge on various logic topics through interactive quizzes. By offering a range of customizable quizzes and immediate feedback, “Quizzeit” aims to make learning formal logic engaging and effective.

Quiz Formal Logical

1. Specific quiz contents

- a. **Propositions and Logical Connectives:** Define and identify propositions, and use of AND, OR, NOT, IF-THEN, IF AND ONLY IF. For example, which logical connective is represented by the symbol $\vee$?
- b. **Truth Tables:** Develop truth tables for given logical expressions, and determine the truth value of complex logical expressions. For example, what is the truth value of $(p \rightarrow q) \wedge (\neg q)$ when p is true and q is false?
- c. **Logical Equivalences:** Recognize or simplify complex logical expressions using logical equivalences. For example, Which of the following is equivalent to $\neg(p \vee q)$?

2. Question generation method

- a. **Random Selection from Question Bank:**
 - **Question Bank Creation:** Develop an extensive question bank for each topic, containing a wide range of questions. Each question entry includes a unique ID, question stem, four answer options (A, B, C, D), and the correct answer.
 - **Random Selection Process:** When a user selects a topic, the system randomly selects 10 questions from the corresponding question bank to generate a quiz.
- b. **Question Template Generation:**
 - **Template Creation:** Design question templates for various topics and define the rules for generating both the questions and the correct answers. For example, a template for a formal logical question might be: "Which of the following is equivalent to $\{(p \vee q)\}$ " Here, $\{ \}$ and $\{ \}$ are placeholders for signs that will be randomly generated.
 - **Variable Insertion:** Randomly generate the values for the variables in the template. For example, if $\{ \}$ is $\neg$ and $\vee$, the generated question would be: "Which of the following is equivalent to $\neg(p \vee q)$ " The correct answer would be $\neg p \wedge \neg q$.
- c. **Generation of Distractor Options:**
 - **Designing Distractor Options:** Establish rules for creating distractor options that differ from the correct answer. For example, generate random formal logical expressions that are close to but not equal to the correct answer, ensuring that there is only one correct answer among the choices.

3. Server-Side Update

- a. **Question Bank Database:** Create comprehensive question bank tables for each topic and difficulty level, storing question ID, question stem, options, and correct answers.

- b. **Question Generation Logic:** Implement the logic for template-based and algorithmic question generation on the server-side to ensure each generated question is unique and has only one correct answer.
- c. **Scoring and Feedback:** When users submit their answers, the server-side scores the answers based on the correct answers and returns the results and correct answers to the users.

App Features

1. User Registration and Authentication Page

- a. **Create an account:** Users can sign up by using their email address and set a password by themselves. The app will send a verification email to users that prevent duplication of accounts.
- b. **Login:** Users can login by using their registered email address, and the app should support password recovery by email.

2. User Profile Page

- a. **Profile Information:** Users can view and edit their profile information, including username, email, and bio. Users can also upload and change their profile images.
- b. **Performance Summary:** Shows a summary of the user's overall performance, including total quizzes taken, average score, and improvement metrics.
- c. **Account Settings:** Users can manage their account settings, including changing their password and setting preferences for notifications.

3. Quiz Working Page

- a. **Question Presentation and Answer Selection:** Each quiz presents a series of questions with multiple-choice answers, and users can navigate questions before submitting their answers.
- b. **Answer Submission:** Users submit their answers for scoring once they complete the quiz. There is a pop up window reminding users to submit.
- c. **Timer and Progress Tracker:** Displays a timer for each quiz and a progress bar showing how many questions have been answered.

4. Quiz List Page

- a. **Quiz List:** Users can select from a list of available quizzes, each with a unique title, description and current score.
- b. **Quiz Ranking:** Users can choose different ranking way to shows all quizzes (e.g Date, Score, Correct Rate)

5. Admin Page

- a. **Overall Performance Display:** Displays total quizzes taken, average score, highest score. Lists all quizzes taken by the user, including dates and scores for each quiz.
- b. **Graphical Representation:** Uses a line chart to indicate score improvements for user's performance trends over time. For example displays the percentage of correct answers across all quizzes taken, helping users identify areas where they need improvement.

Quizzeit

Create an account

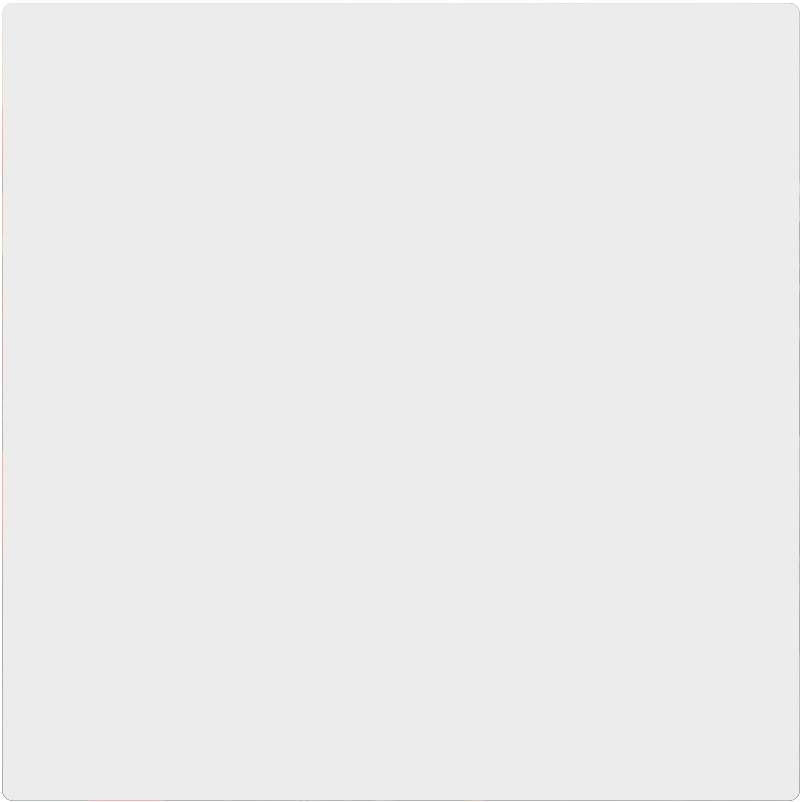
Enter your email to sign up for this app

Sign up with email



Quiz 1

Left Time:



Question #

Specific Question

Please select one answer

A.

B.

C.

D.





Profile Setting



[Edit profile image](#)

Name Admin >

Username @Admin >

Email name@domain.com >

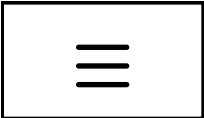
Links
website.net
mylink.net
yourlink.net
[+ Add link](#)

Bio A description of this user. >



Quiz Selection

Grade	Title	
80/100	Quiz 1	Start
60/100	Quiz 2	Start
0/0	Quiz 3	Start



Quizeit

Total Quiz

3

Total Score

2,405

Correct Rate %



Reminder

It is time to start quiz 3!

email@admin.net

